# Statistical Methods

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August 20, 2026

# Outline I

1. Random Variables

2. Standard Distributions

# Why Do We Need Random Variables?

Up to now, we have defined events as qualitative subsets of a sample space:

$$\mathcal{S} = \{\text{Heads}, \text{Tails}\} \quad \text{or} \quad \mathcal{S} = \{\text{Red}, \text{Blue}, \text{Green}\}$$

### The Limitation of Qualitative Outcomes

- We cannot perform algebraic operations (like addition, averaging, or scaling) on raw outcomes like "Heads" or "Red".
- To build mathematical models, calculate averages, or measure variances, we must translate these outcomes into numerical values.

*We need a systematic rule that assigns a real number to every single outcome in our sample space.*

# Formal Definition of a Random Variable

## Definition

Random Variable Let $\mathcal{S}$ be the sample space of a random experiment. A **random variable** $X$ is a real-valued function that maps every outcome $s \in \mathcal{S}$ to a unique real number $X(s) \in \mathbb{R}$.

$$X : \mathcal{S} \to \mathbb{R}$$

## Crucial Conceptual Caveat

Despite its name, a random variable is **not a variable** and is **not random** in the traditional sense.

- It is a deterministic, mathematically defined **function**.
- The "randomness" comes solely from the underlying experiment which determines which outcome $s \in \mathcal{S}$ occurs.

# The Range of a Random Variable

The collection of all numerical values that a random variable can actually take is called its **range** or **support**, often denoted by $\mathcal{T}$ or $X(\mathcal{S})$.

$$\mathcal{T} = \{x \in \mathbb{R} \mid X(s) = x \text{ for some } s \in \mathcal{S}\}$$

Example: Tossing Two Fair Coins

Let the sample space be:

$$\mathcal{S} = \{HH, HT, TH, TT\}$$

Let $X$ be the random variable representing the **number of Heads**.

- $X(HH) = 2$
- $X(HT) = 1$
- $X(TH) = 1$
- $X(TT) = 0$

The support of $X$ is the discrete set: $\mathcal{T} = \{0, 1, 2\}$.

# Discrete vs. Continuous Random Variables

Depending on the nature of their support set $\mathcal{T}$, random variables are classified into two fundamental domains:

## 1. Discrete Random Variables

A random variable is **discrete** if its range $\mathcal{T}$ consists of a finite or countably infinite set of values (such as integers).

- Outcomes can be listed in a sequence: $\{x_1, x_2, x_3, \ldots\}$.

## 2. Continuous Random Variables

A random variable is **continuous** if its range $\mathcal{T}$ consists of an uncountable interval or union of intervals of real numbers.

- It can assume any real-valued measurement in its range.

# Discrete Example: Rolling Two Dice

Suppose two fair, six-sided dice are rolled, yielding a sample space $\mathcal{S}$ of 36 equally likely outcomes.

Let $Y$ be the random variable representing the **sum of the two dice**:

$$Y(i,j) = i + j \quad \text{where } (i,j) \in \mathcal{S}$$

- The minimum possible value is: $Y(1,1) = 2$.
- The maximum possible value is: $Y(6,6) = 12$.
- The range is the discrete list of values:

$$\mathcal{T} = \{2, 3, 4, 5, 6, 7, 8, 9, 10, 11, 12\}$$

Since the range is finite, $Y$ is a **discrete random variable**.

# Continuous Example: Device Lifetime

Consider an experiment where we observe the operational lifetime (in hours) of a continuous electronic component until it experiences a structural failure.

The physical sample space consists of all non-negative real numbers:

$$\mathcal{S} = \{t \in \mathbb{R} \mid t \geq 0\}$$

Let $T$ be the random variable representing the **lifetime of the component**:

$$T(t) = t \quad \text{for any } t \in \mathcal{S}$$

- Since the time of failure can be any fractional value (e.g., 104.532 hours), the range is the continuous interval:

$$\mathcal{T} = [0, \infty)$$

Because this range is uncountable, $T$ is a **continuous random variable**.

# Exercise: Classifying Random Variables

For each of the following scenarios, identify:

1. The random variable $X$ of interest.
2. Whether $X$ is **discrete** or **continuous**.
3. The mathematically defined range/support $\mathcal{T}$ of $X$.

## Scenarios

- **Scenario A:** A fair coin is tossed repeatedly until the first "Heads" appears. We count the total number of tosses required.
- **Scenario B:** A researcher measures the precise blood pressure level (in mmHg) of a patient.
- **Scenario C:** A commuter records the exact waiting time (in minutes) for a city bus, which is guaranteed to arrive within 30 minutes.

# The Induced Probability Map

A random variable $X : \mathcal{S} \to \mathbb{R}$ is a completely deterministic, algebraic function. It does not possess any "inherent" randomness.

### Where does the probability come from?

The probability of $X$ taking on a specific value is entirely **derived** from the probability of the underlying outcomes in the sample space $\mathcal{S}$ that map to that value.

If the underlying physical experiment is performed and yields an outcome $s \in \mathcal{S}$, the random variable evaluates to the number $X(s)$.

# Translating Events to Real Numbers

Let $x$ be a real number. We want to define what $P(X = x)$ means.

We look backward at the sample space $\mathcal{S}$ and gather all outcomes $s$ that map to $x$. We call this subset the **pre-image** of $x$:

$$E_x = \{s \in \mathcal{S} \mid X(s) = x\}$$

### Definition

The Probability Law of $X$ The probability that the random variable $X$ equals $x$ is defined as the probability of its underlying event $E_x$ occurring:

$$P(X = x) \triangleq P(\{s \in \mathcal{S} \mid X(s) = x\})$$

This is often written in compact shorthand as:

$$P(X = x) = P(s \in \mathcal{S} : X(s) = x)$$

# Example: Tossing Two Fair Coins

Let $\mathcal{S} = \{HH, HT, TH, TT\}$ with equally likely outcomes ($P(s) = 1/4$). Let $X$ represent the **number of Heads**:

$$X(HH) = 2, \quad X(HT) = 1, \quad X(TH) = 1, \quad X(TT) = 0$$

To find $P(X = 1)$, we identify which outcomes in $\mathcal{S}$ map to 1:

$$E_1 = \{s \in \mathcal{S} \mid X(s) = 1\} = \{HT, TH\}$$

Since $E_1$ is a standard, well-defined event in our sample space, we can calculate its probability:

$$P(X = 1) = P(E_1) = P(\{HT, TH\}) = \frac{2}{4} = 0.5$$

*The number 1 is assigned the collective probability mass of the outcomes HT and TH.*

# Extending to Intervals and Ranges

We do not just calculate probability at single points. We often want to know the probability of a range, such as $P(a \leq X \leq b)$ or $P(X \leq x)$.

The exact same **mapping** rule applies! We collect all outcomes in the sample space $\mathcal{S}$ that map into that interval:

$$P(a \leq X \leq b) \triangleq P(\{s \in \mathcal{S} \mid a \leq X(s) \leq b\})$$

## Shorthand Notation Caveat

When you write $P(X \leq x)$, you are using a convenient shorthand.
The mathematical reality behind that shorthand is:

$$P(X \leq x) = P(\{s \in \mathcal{S} \mid X(s) \leq x\})$$

This translation connects the real number world $(X \leq x)$ back to the set theory world $(\mathcal{S})$ where Kolmogorov's Axioms live!

# Exercise 1: Cloud Database Reliability

A cloud database replicates data across 3 independent servers. The sample space $\mathcal{S}$ represents the state of each server: $(x_1, x_2, x_3)$ where $x_i \in \{0, 1\}$ ($1 =$ online, $0 =$ offline). The probability of any individual server failing (offline) is 0.1. The servers operate independently.

Let the random variable $C$ represent the **available system throughput** (in Gbps), mapped as a function of the number of online servers $k = x_1 + x_2 + x_3$:

$$C(s) = \begin{cases} 100 & \text{if } k = 3 \\ 50 & \text{if } k = 2 \\ 20 & \text{if } k = 1 \\ 0 & \text{if } k = 0 \end{cases}$$

Calculate the probability $P(C \geq 50)$.

# Exercise 2: Reactor Pressure Telemetry

Consider an industrial pressure sensor placed in a chemical reactor. The operational pressure $s$ (in psi) ranges uniformly over the sample space $\mathcal{S} = [0, 200]$ psi, where $P([a, b]) = \frac{b-a}{200}$ for any sub-interval.
The safety telemetry system converts the raw pressure $s$ to a safety index $Y$ using a piecewise logarithmic mapping:

$$Y(s) = \begin{cases} 0 & \text{if } 0 \leq s < 1 \quad \text{(noise floor)} \\ 20\log_{10}(s) & \text{if } 1 \leq s \leq 100 \quad \text{(telemetry monitoring)} \\ 40 & \text{if } 100 < s \leq 200 \quad \text{(saturation limit)} \end{cases}$$

## Your Task

1. Translate the telemetry reading event $\{20 \leq Y \leq 40\}$ back to its corresponding pre-image subset $E \subseteq \mathcal{S}$.
2. Calculate the probability $P(20 \leq Y \leq 40)$ using the probability law.

# What is a "Distribution"?

A random variable $X$ maps outcomes from a sample space $\mathcal{S}$ to the real numbers $\mathbb{R}$.
The way the probability mass is scattered, allocated, or "distributed" over these real numbers is called the **probability distribution** of $X$.

To study distributions systematically, we use specific functions:

- **The Cumulative Distribution Function (CDF)** — Applies universally to all random variables.
- **The Probability Mass Function (PMF)** — Specifically for discrete random variables.
- **The Probability Density Function (PDF)** — Specifically for continuous random variables.

# Why Do We Need a Distribution Function?

Previously, calculating probabilities required looking backward to the sample space:

$$P(X \in B) = P(\{s \in \mathcal{S} \mid X(s) \in B\})$$

While mathematically sound, this approach has severe practical limitations:

- It is highly inefficient to map back to the sample space $\mathcal{S}$ for every single probability calculation.
- In many real-world applications, the original sample space $\mathcal{S}$ is complex, high-dimensional, or completely hidden from us.

### The Solution

A **distribution function** acts as a mathematical proxy. Once we establish this function on the real line $\mathbb{R}$:

- **All** probabilities associated with $X$ can be calculated using this function **alone**.

# The Cumulative Distribution Function (CDF)

The most fundamental and universal way to describe the distribution of any random variable is through cumulative probabilities.

### Definition

Cumulative Distribution Function Let $X$ be a random variable. The **Cumulative Distribution Function (CDF)** of $X$, denoted by $F(x)$ (or $F_X(x)$), is defined for any real number $x$ as:

$$F(x) \triangleq P(X \leq x) \quad \text{for } x \in \mathbb{R}$$

### Key Concept

The value $F(x)$ represents the accumulated probability of all outcomes in the sample space that map to a value less than or equal to $x$.

# Why $(-\infty, x]$ is Enough: Reconstructing Events

The CDF is defined strictly using semi-infinite intervals:

$$F(a) \triangleq P(X \leq x) = P(X \in (-\infty, x])$$

### A Fundamental Result

Every "nice" or mathematically well-behaved subset of the real numbers can be constructed from intervals of the form $(-\infty, x]$ using basic set operations.

Consequently, the probability of any reasonable event can be written as an addition or subtraction combination of the distribution function values:

# Properties of the CDF

The CDF $F(x)$ is a mathematically well-behaved function. For any random variable $X$, its CDF satisfies the following structural properties:

1. **Monotonicity:** $F(x)$ is a non-decreasing function:

$$\text{If } a \leq b, \text{ then } F(a) \leq F(b)$$

2. **Asymptotic Limits:**

$$\lim_{x \to \infty} F(x) = 1 \quad \text{and} \quad \lim_{x \to -\infty} F(x) = 0$$

3. **Right-Continuity:** $F(x)$ is continuous from the right at every point:

$$\lim_{t \to x^+} F(t) = F(x)$$

# Discrete Case: Probability Mass Function (PMF)

For discrete random variables (which take on countable values), we define probability at exact, individual points.

## Definition

Probability Mass Function Let $X$ be a discrete random variable with range $\mathcal{T}$. The **Probability Mass Function (PMF)** of $X$, denoted by $p(x)$ (or $p_X(x)$), is:

$$p(x) \triangleq P(X = x) \quad \text{for } x \in \mathbb{R}$$

## Fundamental Axiomatic Properties

Every valid PMF must satisfy:

- $p(x) \geq 0$ for all $x$.
- $\sum_{x \in \mathcal{T}} p(x) = 1$.

# Continuous Case: Probability Density Function (PDF)

For continuous random variables, the probability of taking on any single exact value is zero ($P(X = x) = 0$). Instead, we describe the probability over an interval using density.

## Definition

Probability Density Function Let $X$ be a continuous random variable. A non-negative, integrable function $f(x)$ is the **Probability Density Function (PDF)** of $X$ if for any interval $[a, b]$:

$$P(a \leq X \leq b) = \int_a^b f(x)\, dx$$

## Axiomatic Constraints

- $f(x) \geq 0$ for all $x \in \mathbb{R}$.
- $\int_{-\infty}^{\infty} f(x)\, dx = F(\infty) = 1$.

# Exercise: Verifying a Valid PMF

Suppose a system database query returns a variable number of matching records, $X$. The probability mass function of $X$ is modeled by:

$$p(x) = \begin{cases} k \cdot x^2 & \text{if } x \in \{1, 2, 3, 4\} \\ 0 & \text{otherwise} \end{cases}$$

## Your Task

1. Find the value of the normalization constant $k$ such that $p(x)$ represents a mathematically valid PMF.
2. Determine and write down the Cumulative Distribution Function (CDF) $F(x)$ for this discrete random variable.

# Exercise: Verifying a Valid PDF

Consider a continuous random variable $X$ representing the time (in hours) a user spends actively on a platform in a session. The density is modeled by:

$$f(x) = \begin{cases} c \cdot (2x - x^2) & \text{if } x \in [0, 2] \\ 0 & \text{otherwise} \end{cases}$$

## Your Task

Find the value of the normalization constant $c$ such that $f(x)$ represents a mathematically valid PDF, and determine the CDF $F(x)$.

# Expectation: The Center of Mass

The **mathematical expectation** (or mean) of a random variable represents its long-run average value over infinitely repeated trials.

Physically, it acts as the **center of gravity** (centroid) of the probability distribution.

### Definition

Expected Value (Discrete) Let $X$ be a discrete random variable with support set $\mathcal{T}$ and probability mass function $p(x) = P(X = x)$. The expected value of $X$, denoted $E[X]$ or $\mu$, is:

$$E[X] \triangleq \sum_{x \in \mathcal{T}} x \cdot p(x)$$

This sum is defined provided that it is absolutely convergent, i.e., $\sum |x| p(x) < \infty$.

# Expectation: Continuous Analogue

For a continuous random variable, we cannot sum over individual points because any single point has zero probability.
Instead, we integrate over the continuous support using the probability density function (PDF).

### Definition

Expected Value (Continuous) Let $X$ be a continuous random variable with probability density function $f(x)$. The expected value of $X$ is defined as:

$$E[X] \triangleq \int_{-\infty}^{\infty} x \cdot f(x)\, dx$$

This integral is defined provided that it is absolutely convergent, i.e., $\int |x| f(x)\, dx < \infty$.

# Linearity of Expectation

Expectation is a **linear operator**. This means it preserves scaling and addition, regardless of whether the random variables are dependent or independent.

### Theorem

*Linearity Properties Let $X$ be a random variable, and let $a$ and $b$ be constants. Then:*

$$E[aX + b] = aE[X] + b$$

# Proof: Linearity of Expectation

We prove this for the discrete case. The continuous proof follows identically by replacing summations with integrals.
Let $X$ have support $\mathcal{T}$ and PMF $p(x)$. Apply the definition of expectation to $g(X) = aX + b$:

$$\begin{aligned} E[aX+b] &= \sum_{x \in \mathcal{T}} (ax+b) \cdot p(x) \\ &= \sum_{x \in \mathcal{T}} (ax \cdot p(x) + b \cdot p(x)) \\ &= \sum_{x \in \mathcal{T}} ax \cdot p(x) + \sum_{x \in \mathcal{T}} b \cdot p(x) \end{aligned}$$

Pulling the constants out of the summations yields:

$$E[aX+b] = a \sum_{x \in \mathcal{T}} x \cdot p(x) + b \sum_{x \in \mathcal{T}} p(x)$$

Applying $E[X] = \sum xp(x)$ and Axiom 2 ($\sum p(x) = 1$):

$$E[aX+b] = aE[X] + b \cdot (1) = aE[X] + b \quad \blacksquare$$

# Variance: Quantifying Spread

While the mean tells us where the distribution is centered, it tells us nothing about how spread out the values are.

**Variance** measures the average squared deviation of the random variable from its mean ($\mu$).

### Definition

Variance Let $X$ be a random variable with mean $\mu = E[X]$. The variance of $X$, denoted $Var(X)$ or $\sigma^2$, is:

$$Var(X) \triangleq E[(X - \mu)^2]$$

### Standard Deviation

Because variance is measured in squared units, we often use the **Standard Deviation** ($\sigma$) to keep the units consistent with the original data:

# Standard Formula for Variance

Evaluating the definition $E[(X-\mu)^2]$ directly can be algebraically tedious. There exists a much simpler, standard formula.

**Theorem**

*For any random variable $X$:*

$$Var(X) = E[X^2] - (E[X])^2$$

# Derivation:

Let $\mu = E[X]$. Expanding the squared term inside the expectation definition:

$$Var(X) = E[(X-\mu)^2] = E[X^2 - 2\mu X + \mu^2]$$

Now, apply the linearity of expectation to distribute the operator across each term:

$$Var(X) = E[X^2] - E[2\mu X] + E[\mu^2]$$

Since $\mu$ is a constant, we can pull it out of the expectation. Also, the expectation of a constant is the constant itself ($E[\mu^2] = \mu^2$):

$$Var(X) = E[X^2] - 2\mu E[X] + \mu^2$$

Substitute $E[X] = \mu$ back into the expression:

$$Var(X) = E[X^2] - 2\mu(\mu) + \mu^2 = E[X^2] - 2\mu^2 + \mu^2$$

$$Var(X) = E[X^2] - \mu^2 = E[X^2] - (E[X])^2 \quad \blacksquare$$

# Properties of Variance: Scaling and Shifts

Unlike expectation, variance is **not** a linear operator. Adding a constant shift does not change the spread of the data, while scaling a variable scales its variance quadratically.

### Theorem

*Variance Transformations Let $X$ be a random variable, and let $a$ and $b$ be constants. Then:*

$$Var(aX + b) = a^2 Var(X)$$

# Proof: Variance Transformations

Let $Y = aX + b$. First, find the mean of the transformed variable $Y$:

$$E[Y] = E[aX + b] = aE[X] + b$$

Now, substitute $Y$ and $E[Y]$ into the definition of variance:

$$\begin{aligned} Var(aX + b) &= E\left[((aX + b) - E[aX + b])^2\right] \\ &= E\left[(aX + b - (aE[X] + b))^2\right] \end{aligned}$$

Notice that the constant shift $b$ cancels out completely:

$$\begin{aligned} Var(aX + b) &= E\left[(aX - aE[X])^2\right] \\ &= E\left[(a(X - E[X]))^2\right] \\ &= E\left[a^2(X - E[X])^2\right] \end{aligned}$$

Pulling the constant factor $a^2$ out of the expectation:

$$Var(aX + b) = a^2 E\left[(X - E[X])^2\right] = a^2 Var(X) \quad \blacksquare$$

# Exercise 1: Server Latency Cost Mapping

An API server experiences variable request latency. Let $X$ represent the processing latency (in milliseconds) with the following PMF:

| $x$ | 10 ms | 50 ms | 100 ms |
|---|---|---|---|
| $p(x)$ | 0.6 | 0.3 | 0.1 |

The operating cost (in micro-dollars) of processing a request of latency $X$ is determined by the quadratic relationship:

$$C(X) = 2X^2 + 10$$

### Your Task

Calculate the expected processing cost $E[C(X)]$ per request.

# Exercise 2: Precision Sensor Noise

An industrial analog temperature sensor outputs a voltage reading $V$. Due to thermal interference, the sensor error $X = V - V_{\text{true}}$ (in millivolts) is a continuous random variable distributed uniformly on the range $[-5, 5]$ mV, with a constant probability density:

$$f(x) = \frac{1}{10} \quad \text{for } x \in [-5, 5]$$

## Your Task

1. Compute the expected calibration error $E[X]$.
2. Compute the variance $Var(X)$ to quantify the sensor's precision limit.

# Central Tendency: The Mathematical Mean

The expected value $E[X]$ we defined is physically known as the **mean** ($\mu$). It is the most common measure of **central tendency**.

## Key Characteristics of the Mean

- **Physical Centroid:** It acts as the exact balancing point of the probability mass curve or histogram.
- **Sensitivity:** Because its calculation factors in every value weighted by its probability, the mean is **highly sensitive to outliers** (skewed tails).
- **Linear Balance:** The sum (or integral) of deviations from the mean is always exactly zero:

$$E[X - \mu] = 0$$

# Alternative Measures: The Median

When a distribution has extreme values or heavy asymmetry, the mean may not reflect the "typical" value. In such cases, we use the **median**.

### Definition

The Median ($m$) The median $m$ of a random variable $X$ is a real number that divides the probability distribution into two equal halves.

- **Continuous Case:** $m$ is the value satisfying:

$$F(m) = \int_{-\infty}^{m} f(x)\, dx = 0.5$$

- **Discrete Case:** $m$ is any value satisfying:

$$P(X \leq m) \geq 0.5 \quad \text{and} \quad P(X \geq m) \geq 0.5$$

**Core Advantage:** The median is **robust**; changing extreme values has absolutely zero effect on its position.

# Alternative Measures: The Mode

Another fundamental measure of central tendency is the **mode**, which indicates the most likely or popular outcome.

### Definition

The Mode ($x^*$) The mode $x^*$ is the value of $x$ where the probability distribution reaches its maximum concentration or peak.

- **Discrete Case:** The point $x^*$ that maximizes the PMF:

$$p(x^*) \geq p(x) \quad \text{for all } x$$

- **Continuous Case:** The point $x^*$ that maximizes the density function $f(x)$ (typically found where $f'(x^*) = 0$ and $f''(x^*) < 0$).

### Distribution Shapes

Unlike mean and median, a distribution can be **unimodal** (one peak), **bimodal** (two peaks), or **multimodal** (several peaks).

# Comparing Central Tendency: Symmetry vs. Skewness

The relative positions of the Mean, Median, and Mode reveal the geometric **skewness** (asymmetry) of a distribution.

## 1. Perfectly Symmetric Distributions

The peaks, physical balance points, and 50% splitting lines align perfectly:

$$\text{Mean} = \text{Median} = \text{Mode}$$

## 2. Skewed Distributions

- **Right-Skewed (Positive):** A long right tail pulls the mean upward. The mode remains at the main peak, with the median sitting safely in between:

$$\text{Mode} < \text{Median} < \text{Mean}$$

- **Left-Skewed (Negative):** A long left tail pulls the mean downward:

$$\text{Mean} < \text{Median} < \text{Mode}$$

# Robust Variability: Median Absolute Deviation

Just as the mean has a robust counterpart (the median), variance has a robust alternative known as the **Median Absolute Deviation (MAD)**. Variance and standard deviation are highly sensitive to extreme outliers because they square deviations. MAD resolves this.

### Definition

Median Absolute Deviation (MAD) Let $X$ be a random variable with median $m$. The Median Absolute Deviation of $X$ is defined as:

$$MAD \triangleq \text{median}(|X - m|)$$

### Why Use MAD?

- **Extreme Outlier Resistance:** A single outlier can inflate variance infinitely, whereas it barely changes the MAD.
- **Preserves Scale:** It measures variability in the same units as the original data without squaring the errors.

# Why Do We Need Joint Distributions?

In most real-world scenarios, an experiment produces multiple numerical outcomes simultaneously.
To model relationships, associations, and dependencies, we must look at variables collectively rather than in isolation.

## Key Objectives

- Learn to describe the joint behavior of multiple random variables.
- Understand how to extract individual probability laws from joint laws.
- Quantify the strength and direction of linear relationships between variables.

# Discrete Case: Joint Probability Mass Function

When $X$ and $Y$ are both discrete random variables, we describe their joint probabilities using a single function of two variables.

### Definition

Joint Probability Mass Function The joint probability mass function (joint PMF) of discrete random variables $X$ and $Y$ is defined for all real numbers $x$ and $y$ as:

$$p(x, y) \triangleq P(X = x, Y = y)$$

### Axiomatic Properties

Every valid joint PMF must satisfy:

- $p(x, y) \geq 0$ for all $(x, y)$.
- $\sum_x \sum_y p(x, y) = 1$.

# Continuous Case: Joint Probability Density Function

When variables are continuous, probability at a single point is zero. Instead, we define probability over regions using a joint density surface.

## Definition

Joint Probability Density Function The joint probability density function (joint PDF) of continuous random variables $X$ and $Y$ is a function $f(x, y)$ such that for any region $A \subseteq \mathbb{R}^2$:

$$P((X, Y) \in A) = \iint_A f(x, y)\, dx\, dy$$

## Required Constraints

- $f(x, y) \geq 0$ for all $(x, y) \in \mathbb{R}^2$.
- $\int_{-\infty}^{\infty} \int_{-\infty}^{\infty} f(x, y)\, dx\, dy = 1$.

# Extracting Individual Laws: Marginal Distributions

If we are given a joint distribution, we can recover the individual probability law of $X$ or $Y$ by summing or integrating out the other variable.

### Marginal PMF (Discrete)

We sum across all possible values of the unwanted variable:

$$p_X(x) = \sum_y p(x, y) \quad \text{and} \quad p_Y(y) = \sum_x p(x, y)$$

### Marginal PDF (Continuous)

We integrate across the entire domain of the unwanted variable:

$$f_X(x) = \int_{-\infty}^{\infty} f(x, y)\, dy \quad \text{and} \quad f_Y(y) = \int_{-\infty}^{\infty} f(x, y)\, dx$$

# Example: Discrete Joint PMF

Let $X$ and $Y$ be discrete random variables with the following joint PMF table:

| $X \setminus Y$ | $y = 0$ | $y = 1$ | $y = 2$ |
|---|---|---|---|
| $x = 0$ | 0.1 | 0.2 | 0.1 |
| $x = 1$ | 0.2 | 0.3 | 0.1 |

## Finding the Marginal PMF of $X$

To find $p_X(x)$, we sum across the rows (integrating out $Y$):

- $p_X(0) = \sum_{y=0}^{2} p(0, y) = 0.1 + 0.2 + 0.1 = 0.4$
- $p_X(1) = \sum_{y=0}^{2} p(1, y) = 0.2 + 0.3 + 0.1 = 0.6$

*Sanity Check:* $\sum_x p_X(x) = 0.4 + 0.6 = 1.$

# Example: Continuous Joint PDF

Let $X$ and $Y$ have the following joint density function:

$$f(x, y) = \begin{cases} c(x + 2y), & 0 < x < 1,\ 0 < y < 1 \\ 0, & \text{otherwise} \end{cases}$$

Step 1: Find the normalizing constant $c$

We use the total probability constraint $\int \int f(x, y)\, dx\, dy = 1$:

$$\begin{aligned} 1 &= \int_0^1 \int_0^1 c(x + 2y)\, dx\, dy \\ &= c \int_0^1 \left[\frac{x^2}{2} + 2xy\right]_{x=0}^{x=1} dy \\ &= c \int_0^1 \left(\frac{1}{2} + 2y\right) dy \\ &= c \left[\frac{y}{2} + y^2\right]_0^1 = c\left(\frac{1}{2} + 1\right) = \frac{3}{2}c \end{aligned}$$

## Example: Continuous Marginal PDF

Continuing with our normalized joint PDF: $f(x, y) = \frac{2}{3}(x + 2y)$ for $0 < x < 1$ and $0 < y < 1$.

### Step 2: Find the marginal PDF of $X$

To find $f_X(x)$, we integrate out $Y$ over its entire domain:

$$\begin{aligned} f_X(x) &= \int_{-\infty}^{\infty} f(x, y)\, dy \\ &= \int_0^1 \frac{2}{3}(x + 2y)\, dy \\ &= \frac{2}{3}\left[xy + y^2\right]_{y=0}^{y=1} \\ &= \frac{2}{3}(x + 1) \end{aligned}$$

Therefore, $f_X(x) = \frac{2}{3}(x + 1)$ for $0 < x < 1$, and 0 otherwise.

# Independence of Random Variables

We previously defined independence for events. Similarly, two random variables are independent if knowing the value of one provides zero information about the other.

### Definition

Statistical Independence Random variables $X$ and $Y$ are independent if their joint distribution factors cleanly into the product of their individual marginal distributions.

- **Discrete Case:**

$$p(x, y) = p_X(x) \cdot p_Y(y) \quad \text{for all } x, y$$

- **Continuous Case:**

$$f(x, y) = f_X(x) \cdot f_Y(y) \quad \text{for all } x, y$$

# Linearity of Expectation

Expectation is a **linear operator**. This means it preserves scaling and addition, regardless of whether the random variables are dependent or independent.

### Theorem

*Linearity Properties Let $X$ and $Y$ be two jointly distributed random variables. Then:*

$$E[X+Y] = E[X] + E[Y]$$

# Proof: Linearity of Expectation

To fully establish linearity, we must show that expectation distributes over addition for **any** two random variables $X$ and $Y$, without assuming independence. Let $X$ and $Y$ have joint support $\mathcal{T}_{X,Y}$ and joint PMF $p(x,y)$. Apply the definition of expectation to $g(X,Y) = X + Y$:

$$E[X+Y] = \sum_{(x,y)\in\mathcal{T}_{X,Y}} (x+y)\cdot p(x,y)$$
$$= \sum_x \sum_y (x\cdot p(x,y) + y\cdot p(x,y))$$
$$= \sum_x \sum_y x\cdot p(x,y) + \sum_y \sum_x y\cdot p(x,y)$$

Factoring out the variables that do not depend on the inner summation index:

$$E[X+Y] = \sum_x x\left(\sum_y p(x,y)\right) + \sum_y y\left(\sum_x p(x,y)\right)$$

$$E[X+Y] = \sum_x x\cdot p_X(x) + \sum_y y\cdot p_Y(y) = E[X] + E[Y] \quad \blacksquare$$

# Covariance: Quantifying Relationship Direction

While expectation tells us about the center of individual variables, we need a metric to evaluate how they vary together.

## Definition

Covariance Let $X$ and $Y$ be random variables with means $\mu_X = E[X]$ and $\mu_Y = E[Y]$. The covariance of $X$ and $Y$, denoted $Cov(X, Y)$, is:

$$Cov(X, Y) \triangleq E[(X - \mu_X)(Y - \mu_Y)]$$

## Intuitive Interpretation

- **Positive Covariance:** When $X$ is above its mean, $Y$ tends to be above its mean (positive linear trend).
- **Negative Covariance:** When $X$ is above its mean, $Y$ tends to be below its mean (inverse linear trend).
- **Zero Covariance:** No linear association.

# Covariance

Evaluating the definition $E[(X - \mu_X)(Y - \mu_Y)]$ directly requires calculating deviations for every point, which is algebraically tedious. Instead, we use a much simpler standard identity.

**Theorem**

*For any two random variables $X$ and $Y$:*

$$Cov(X, Y) = E[XY] - E[X]E[Y]$$

*Let us derive this relationship on the next slide using the properties of expectation.*

# Derivation:

Let $\mu_X = E[X]$ and $\mu_Y = E[Y]$. Expanding the terms inside the expectation definition:

$$Cov(X, Y) = E[(X - \mu_X)(Y - \mu_Y)]$$

$$Cov(X, Y) = E[XY - \mu_Y X - \mu_X Y + \mu_X \mu_Y]$$

Apply the linearity of expectation to distribute the operator across each term:

$$Cov(X, Y) = E[XY] - E[\mu_Y X] - E[\mu_X Y] + E[\mu_X \mu_Y]$$

Since the means $\mu_X$ and $\mu_Y$ are constants, we pull them out of the expectation:

$$Cov(X, Y) = E[XY] - \mu_Y E[X] - \mu_X E[Y] + \mu_X \mu_Y$$

Substitute $E[X] = \mu_X$ and $E[Y] = \mu_Y$ back into the expression:

$$Cov(X, Y) = E[XY] - \mu_Y \mu_X - \mu_X \mu_Y + \mu_X \mu_Y$$

$$Cov(X, Y) = E[XY] - \mu_X \mu_Y = E[XY] - E[X]E[Y] \quad \blacksquare$$

# Properties of Covariance

Covariance behaves according to the following mathematical rules:

## Core Covariance Properties

Let $X$, $Y$, and $Z$ be random variables, and let $a$ and $b$ be constants.

1. **Symmetry:** $Cov(X, Y) = Cov(Y, X)$
2. **Covariance with Self is Variance:** $Cov(X, X) = Var(X)$
3. **Scaling and Bilinearity:** $Cov(aX, bY) = ab\,Cov(X, Y)$
4. **Additivity:** $Cov(X + Y, Z) = Cov(X, Z) + Cov(Y, Z)$

# Variance is NOT Linear

Unlike expectation, variance is a **quadratic** operator. Because it measures squared deviations, it does not simply preserve addition or scaling.

### The General Rule for Variance

Let $X$ and $Y$ be any two random variables, and let $a, b \in \mathbb{R}$ be constants. The variance of their linear combination is:

$$Var(aX + bY) = a^2 Var(X) + b^2 Var(Y) + 2ab\, Cov(X, Y)$$

If we simply add the variables together ($a = 1, b = 1$), this reduces to the fundamental sum formula:

$$Var(X + Y) = Var(X) + Var(Y) + 2Cov(X, Y)$$

**Key Takeaway:** $Var(X + Y) \neq Var(X) + Var(Y)$ unless the variables are completely uncorrelated (i.e., $Cov(X, Y) = 0$).

# Proof: Variance of a Sum

Let us formally prove that $Var(X+Y) = Var(X) + Var(Y) + 2Cov(X,Y)$. By definition, $Var(Z) = E[(Z - E[Z])^2]$. Let $Z = X + Y$. Applying our rule of linear expectation, $E[X+Y] = E[X] + E[Y]$.

## Algebraic Expansion

Group the random variables directly with their respective population means:

$$\begin{aligned} Var(X+Y) &= E\left[\Big((X+Y) - (E[X] + E[Y])\Big)^2\right] \\ &= E\left[\Big((X - E[X]) + (Y - E[Y])\Big)^2\right] \end{aligned}$$

Expand to get:

$$= E\left[(X - E[X])^2 + 2(X - E[X])(Y - E[Y]) + (Y - E[Y])^2\right]$$

$$\implies Var(X+Y) = Var(X) + 2Cov(X,Y) + Var(Y)$$

# Independence and Covariance

What happens to covariance if the random variables are independent?

### Theorem

*Covariance of Independent Variables If $X$ and $Y$ are independent random variables, then:*

$$Cov(X, Y) = 0$$

### Proof Sketch

If $X$ and $Y$ are independent, then their joint expectation factors: $E[XY] = E[X]E[Y]$. Substituting this into our computational formula:

$$Cov(X, Y) = E[XY] - E[X]E[Y] = E[X]E[Y] - E[X]E[Y] = 0 \quad \blacksquare$$

### The Converse is not true

The converse is **not** generally true. If $Cov(X, Y) = 0$, the variables are uncorrelated, but they are **not necessarily independent**.

# Exercise 1: Discrete Covariance Calculation

Let $X$ and $Y$ be discrete random variables representing the outcome of a simple system. Their joint PMF, $p(x, y)$, is defined by the following table:

| | $Y = 1$ | $Y = 2$ |
|---|---|---|
| $X = 0$ | 0.1 | 0.4 |
| $X = 1$ | 0.3 | 0.2 |

## Your Task

1. Find the marginal PMFs of $X$ and $Y$.
2. Compute the expected values $E[X]$ and $E[Y]$.
3. Compute the covariance $Cov(X, Y)$.

# Exercise 2: Continuous Covariance Calculation

Let $X$ and $Y$ be continuous random variables. Their joint density function is defined as:

$$f(x,y) = \begin{cases} x+y & \text{if } 0 \le x \le 1 \text{ and } 0 \le y \le 1 \\ 0 & \text{otherwise} \end{cases}$$

## Your Task

1. Compute the expected values $E[X]$ and $E[Y]$.
2. Compute the expected product $E[XY]$.
3. Compute the covariance $Cov(X, Y)$.

# The Cauchy-Schwarz Inequality

The relationship between the expectations of products and individual second moments is bounded by a fundamental mathematical law.

### Theorem

*Cauchy-Schwarz Inequality For any two random variables $X$ and $Y$ with finite variances:*

$$(E[XY])^2 \leq E[X^2]E[Y^2]$$

*In terms of deviations from their respective means, this is equivalent to:*

$$(Cov(X, Y))^2 \leq Var(X)Var(Y)$$

### Physical Implication

This inequality guarantees that covariance cannot grow infinitely large. It is bounded strictly by the individual spreads (variances) of the two variables, which allows us to define a standardized scale for relationships.

# Sketch of Proof

The proof relies on the property that the expectation of a non-negative random variable is always non-negative.

### 1. Starting Point

For any constant $a \in \mathbb{R}$, consider the squared difference $E[(aX - bY)^2] \geq 0$.

### 2. Expansion

Expanding the quadratic inside the expectation:

$$E[a^2X^2 - 2aXY + Y^2] \geq 0$$

By the linearity of expectation, we obtain:

$$a^2E[X^2] - 2aE[XY] + E[Y^2] \geq 0$$

# Sketch of Proof

### 3. The Discriminant Condition

Treating this as a quadratic in $a$ , for the expression to be $\geq 0$ for all $a$, the discriminant must be non-positive ($\Delta \leq 0$):

$$(2E[XY])^2 - 4E[X^2]E[Y^2] \leq 0$$

$$4(E[XY])^2 \leq 4E[X^2]E[Y^2]$$

### 4. Result

Dividing by 4 yields the Cauchy-Schwarz Inequality:

$$(E[XY])^2 \leq E[X^2]E[Y^2]$$

### 5. Equality Condition

The equality holds iff there exists an $a$ such that $aX = Y$ wp 1.

# The Correlation Coefficient: Standardizing Covariance

While covariance tells us the direction of a linear relationship, its numerical value depends on the arbitrary units of $X$ and $Y$. To fix this, we normalize it.

## Definition

Correlation Coefficient The correlation coefficient of two random variables $X$ and $Y$, denoted by $\rho(X, Y)$ (or simply $\rho$), is defined as:

$$\rho(X, Y) \triangleq \frac{Cov(X, Y)}{\sigma_X \sigma_Y} = \frac{Cov(X, Y)}{\sqrt{Var(X)Var(Y)}}$$

## Properties of Correlation

By direct application of the Cauchy-Schwarz inequality:

- **Boundedness:** $-1 \leq \rho(X, Y) \leq 1$.
- **Perfect Linear Relationship:** $\rho = 1$ indicates a perfect positive linear trend; $\rho = -1$ indicates a perfect negative linear trend.

# Skewness

For a continuous random variable $X$ with PDF $f(x)$, mean $\mu$, and standard deviation $\sigma$, the skewness is defined via the integral:

$$\gamma_1 = \frac{1}{\sigma^3} \int_{-\infty}^{\infty} (x-\mu)^3 f(x)\, dx$$

## Proof of Zero Skewness for Symmetry

Assume $f(x)$ is perfectly symmetric about $\mu$, meaning $f(\mu - y) = f(\mu + y)$. We can shift our axis by letting $y = x - \mu$:

$$\int_{-\infty}^{\infty} y^3 f(\mu + y)\, dy$$

Notice that $y^3$ is an **odd** function ($(-y)^3 = -y^3$), while our shifted PDF $f(\mu + y)$ is an **even** function. The product of an odd and even function is odd. Integrating an odd function over a symmetric interval $[-\infty, \infty]$ perfectly cancels out, yielding 0.

# Rethinking Kurtosis: The Variance of Deviations

To understand why thicker tails increase kurtosis, let us standardize our variable: $Z = \frac{X-\mu}{\sigma}$. By definition, $E[Z^2] = 1$ and Kurtosis is $E[Z^4]$.Let us examine the variance of these squared deviations, $Var(Z^2)$, using the standard variance expansion $Var(Y) = E[Y^2] - (E[Y])^2$:

$$Var(Z^2) = E[(Z^2)^2] - (E[Z^2])^2$$

$$Var(Z^2) = E[Z^4] - (1)^2$$

Rearranging this gives us some insight into Kurtosis:

$$\text{Kurtosis} = Var(Z^2) + 1$$

It shows that Kurtosis is entirely driven by how much the **squared deviations vary**.

# Why Thicker Tails Drive Kurtosis Higher

The relationship Kurtosis $= 1 + Var(Z^2)$ perfectly explains the geometric behavior of the tails.

## 1. Thick Tails

$Z^2$ represents the squared distance of a point from the mean. If a distribution has **thick tails**, it possesses a higher probability of generating large values producing occasionally massive values for $Z^2$.

## 2. Conclusion

Thicker tails $\implies$ larger extreme $Z^2$ values $\implies$ a massive increase in $Var(Z^2)$. Because Kurtosis is simply $Var(Z^2) + 1$, adding mass to the tails forces the kurtosis higher.

# Standard Discrete Distributions: The Discrete Uniform

We begin our exploration of standard distributions with the most fundamental concept of fairness: a system where every outcome in a finite set is equally likely.

### The Discrete Uniform Distribution

Let a random variable $X$ take on any integer value in a contiguous range from $a$ to $b$ inclusive.

- The total number of possible outcomes is $N = b - a + 1$.
- Because no outcome is favored, the probability is distributed evenly across the entire support.

### Probability Mass Function (PMF)

We write $X \sim \text{DiscreteUniform}(a, b)$. The PMF is:

$$P(X = k) = \frac{1}{N} = \frac{1}{b - a + 1}, \quad \text{for } k = a, a + 1, \ldots, b$$

# Discrete Uniform: Expectation and Variance

### 1. Expectation (The Midpoint)

Using the definition of expectation and factoring out the constant probability:

$$E[X] = \sum_{k=a}^{b} k \frac{1}{N} = \frac{1}{N} \sum_{k=a}^{b} k$$

The sum of this arithmetic sequence is the average of the first and last terms multiplied by the total number of terms ($N$):

$$E[X] = \frac{1}{N} \left( N \frac{a+b}{2} \right) = \frac{a+b}{2}$$

# Discrete Uniform Variance

Calculating the sum of squares directly from $a$ to $b$ is messy. Instead, we define a shifted variable $Y = X - a + 1$.
$Y$ now takes values $1, 2, \ldots, N$. Because shifting a distribution does not change its spread, $Var(X) = Var(Y)$.

## 1. Find the Second Moment $E[Y^2]$

Using the standard formula for the sum of consecutive squares:

$$E[Y^2] = \sum_{k=1}^{N} k^2 \frac{1}{N} = \frac{1}{N}\left[\frac{N(N+1)(2N+1)}{6}\right] = \frac{(N+1)(2N+1)}{6}$$

## 2. Calculate Final Variance

$$Var(Y) = \frac{(N+1)(2N+1)}{6} - \left(\frac{N+1}{2}\right)^2$$

$$\implies Var(Y) = \frac{N+1}{2}\left[\frac{N-1}{6}\right] = \frac{N^2-1}{12} \quad \blacksquare$$

# The Bernoulli Trial

## The Bernoulli Trial

A Bernoulli trial is a random experiment with only two possible outcomes, typically labeled as "Success" (1) and "Failure" (0).

- Let $p$ be the probability of success, where $0 \leq p \leq 1$.
- The probability of failure is therefore $1 - p$ (often denoted as $q$).

**Examples:**

- Flipping a biased coin (Heads $= 1$, Tails $= 0$).
- Testing a manufactured component (Pass $= 1$, Defective $= 0$).
- A bit transmitted over a noisy channel (Received correctly $= 1$, Error $= 0$).

# The Bernoulli Distribution

If a random variable $X$ represents the outcome of a Bernoulli trial, we write $X \sim \text{Bernoulli}(p)$.
The PMF can be compactly written for $x \in \{0, 1\}$ as:

$$P(X = x) = p^x(1 - p)^{1-x}$$

Using our fundamental definitions of expectation and variance:

- **Expectation:**

$$E[X] = \sum_{x\in\{0,1\}} x \cdot P(X = x) = (1 \cdot p) + (0 \cdot (1 - p)) = p$$

- **Second Moment:**

$$E[X^2] = (1^2 \cdot p) + (0^2 \cdot (1 - p)) = p$$

- **Variance:**

$$Var(X) = E[X^2] - (E[X])^2 = p - p^2 = p(1 - p)$$

# Standard Discrete Models: The Binomial Distribution

What happens if we repeat a Bernoulli trial multiple times?

### Definition via Summation

Let $X_1, X_2, \ldots, X_n$ be $n$ **independent** random variables, where each $X_i \sim \text{Bernoulli}(p)$.
We define a new random variable $Y$ as their sum:

$$Y = \sum_{i=1}^{n} X_i$$

$Y$ represents the total number of "Successes" in $n$ independent trials. We say $Y \sim \text{Binomial}(n, p)$.

**Examples:**

- The number of Heads in 10 coin flips.
- The number of defective items in a batch of 50.

# Binomial Probability Mass Function

To find the PMF of $Y$, we want the probability of getting exactly $k$ successes (where $k \in \{0, 1, \dots, n\}$).

## Deriving the Probability

1. **Specific Sequence:** The probability of getting $k$ successes followed exactly by $n-k$ failures is $p^k(1-p)^{n-k}$.
2. **Counting Combinations:** There are many ways to arrange $k$ successes among $n$ trials. The number of such unique sequences is given by the binomial coefficient $\binom{n}{k}$.

Multiplying these together yields the PMF:

$$P(Y = k) = \binom{n}{k} p^k (1-p)^{n-k}$$

# Binomial Mean and Variance

We could calculate the expectation by solving $\sum_{k=0}^{n} k\binom{n}{k}p^k(1-p)^{n-k}$, which requires complex algebra.

Instead, we use the linear properties we proved earlier!

1. Expectation (Linearity)

Because expectation is a linear operator:

$$E[Y] = E\left[\sum_{i=1}^{n} X_i\right] = \sum_{i=1}^{n} E[X_i] = \sum_{i=1}^{n} p = np$$

2. Variance (Independence)

Because the trials $X_i$ are **independent**, the covariance terms vanish, allowing us to distribute the variance operator:

$$Var(Y) = Var\left(\sum_{i=1}^{n} X_i\right) = \sum_{i=1}^{n} Var(X_i) = \sum_{i=1}^{n} p(1-p) = np(1-p)$$

# Binomial Expectation:

Let $Y \sim \text{Binomial}(n, p)$ and $q = 1 - p$. We calculate $E[Y]$ directly:

$$E[Y] = \sum_{k=0}^{n} k \binom{n}{k} p^k q^{n-k} = \sum_{k=1}^{n} k \frac{n!}{k!(n-k)!} p^k q^{n-k}$$

Cancel $k$ with $k!$, leaving $(k-1)!$ in the denominator:

$$E[Y] = \sum_{k=1}^{n} \frac{n!}{(k-1)!(n-k)!} p^k q^{n-k}$$

Factor out $np$ to reconstruct a new binomial coefficient:

$$E[Y] = np \sum_{k=1}^{n} \frac{(n-1)!}{(k-1)!(n-k)!} p^{k-1} q^{n-k} = np \sum_{k=1}^{n} \binom{n-1}{k-1} p^{k-1} q^{(n-1)-(k-1)}$$

Let $j = k - 1$. By the Binomial Theorem, the remaining sum is $(p+q)^{n-1} = 1$:

$$E[Y] = np \cdot 1 = np \quad \blacksquare$$

# Binomial Variance: Second Factorial Moment

To find the variance, we first calculate the **second factorial moment**, $E[Y(Y-1)]$:

$$E[Y(Y-1)] = \sum_{k=0}^{n} k(k-1)\binom{n}{k} p^k q^{n-k}$$

The terms for $k = 0$ and $k = 1$ are zero. Starting at $k = 2$, cancel $k(k-1)$ with $k!$:

$$E[Y(Y-1)] = \sum_{k=2}^{n} \frac{n!}{(k-2)!(n-k)!} p^k q^{n-k}$$

Factor out $n(n-1)p^2$ to rebuild the combinations:

$$E[Y(Y-1)] = n(n-1)p^2 \sum_{k=2}^{n} \frac{(n-2)!}{(k-2)!(n-k)!} p^{k-2} q^{n-k}$$

Just as before, the remaining sum is exactly $(p+q)^{n-2} = 1$. Thus:

$$E[Y(Y-1)] = n(n-1)p^2$$

# Binomial Variance:

We just found $E[Y(Y-1)] = n(n-1)p^2$ and we know $E[Y] = np$.

## 1. Find the Second Moment $E[Y^2]$

Expand the factorial moment: $E[Y(Y-1)] = E[Y^2 - Y] = E[Y^2] - E[Y]$.

Rearranging gives:

$$\begin{aligned} E[Y^2] &= E[Y(Y-1)] + E[Y] \\ &= n(n-1)p^2 + np \\ &= n^2p^2 - np^2 + np \end{aligned}$$

## 2. Calculate Final Variance

Apply the standard variance formula $Var(Y) = E[Y^2] - (E[Y])^2$:

$$\begin{aligned} Var(Y) &= (n^2p^2 - np^2 + np) - (np)^2 \\ &= n^2p^2 - np^2 + np - n^2p^2 \\ &= np - np^2 = np(1-p) \quad \blacksquare \end{aligned}$$

# The Sample Proportion ($\hat{p}$)

Instead of looking at the total successes $Y$, let's examine the **average number of successes** per trial, known as the sample proportion: $\hat{p} = \frac{Y}{n}$.

## 1. Expectation

$$E\left[\frac{Y}{n}\right] = \frac{1}{n}E[Y] = \frac{1}{n}(np) = p$$

The expected proportion of successes is exactly the probability of a single success.

## 2. Variance

Recall that pulling a constant out of a variance operator squares the constant:

$$Var\left(\frac{Y}{n}\right) = \frac{1}{n^2}Var(Y) = \frac{1}{n^2}np(1-p) = \frac{p(1-p)}{n}$$

# The Sample Proportion: As $n \to \infty$

We just established that the variance of the sample proportion is inversely proportional to the number of trials:

$$Var(\hat{p}) = \frac{p(1-p)}{n}$$

## The Infinite Limit

What happens mathematically if we repeat the trial an infinite number of times?

$$\lim_{n\to\infty} Var(\hat{p}) = \lim_{n\to\infty} \frac{p(1-p)}{n} = 0$$

**Physical Implication:** As $n$ grows indefinitely, the sample proportion $\hat{p}$ stops fluctuating and collapses precisely onto the true, underlying probability $p$. This validates the frequentist definition of probability.

# The "Rare Event" Limit

What happens to a Binomial distribution when the number of trials $n$ becomes extremely large, but the probability of success $p$ is extremely small?

## The Setup

Consider modeling rare events in a massive population, such as:

- The number of typos in a 500-page book.
- The number of radioactive decays in a millisecond.

Let us take the limit as $n \to \infty$ and $p \to 0$. To keep the mathematics meaningful, we hold their product constant: let $np = \lambda$.
Here, $\lambda$ represents the **average rate** of successes.

# Deriving the Poisson PMF: Setup

Let $Y \sim \text{Binomial}(n, p)$. We substitute $p = \frac{\lambda}{n}$ into the Binomial Probability Mass Function and take the limit as $n \to \infty$:

$$\lim_{n\to\infty} P(Y = k) = \lim_{n\to\infty} \binom{n}{k} p^k (1-p)^{n-k}$$

$$= \lim_{n\to\infty} \frac{n!}{k!(n-k)!} \left(\frac{\lambda}{n}\right)^k \left(1 - \frac{\lambda}{n}\right)^{n-k}$$

We can group the constants (with respect to $n$) and pull them to the front:

$$= \frac{\lambda^k}{k!} \lim_{n\to\infty} \left[ \frac{n!}{(n-k)!} \cdot \frac{1}{n^k} \cdot \left(1 - \frac{\lambda}{n}\right)^{n-k} \right]$$

# Deriving the Poisson PMF: Evaluation

Let's break the expression inside the limit into three distinct components:

$$= \frac{\lambda^k}{k!} \lim_{n\to\infty} \underbrace{\left[\frac{n(n-1)\dots(n-k+1)}{n^k}\right]}_{\text{Part A}} \underbrace{\left(1-\frac{\lambda}{n}\right)^n}_{\text{Part B}} \underbrace{\left(1-\frac{\lambda}{n}\right)^{-k}}_{\text{Part C}}$$

1. **Part A:** There are exactly $k$ terms in the numerator polynomial and $n^k$ in the denominator. As $n \to \infty$, the highest degree terms dominate, and this fraction approaches 1.
2. **Part B:** From standard calculus, the limit definition of the exponential function dictates that $\lim_{n\to\infty} \left(1 - \frac{\lambda}{n}\right)^n = e^{-\lambda}$.
3. **Part C:** For any finite $k$, as $n \to \infty$, the term inside the parentheses approaches 1. Thus, $1^{-k} = 1$.

# The Poisson Distribution

Multiplying our evaluated limits together $(1 \cdot e^{-\lambda} \cdot 1)$, we arrive at the PMF for a new, fundamental distribution.

## The Poisson PMF

A random variable $X$ follows a Poisson distribution with rate parameter $\lambda > 0$, denoted $X \sim \text{Poisson}(\lambda)$, if:

$$P(X = k) = \frac{\lambda^k e^{-\lambda}}{k!}, \quad \text{for } k = 0, 1, 2, \ldots$$

## Mean and Variance of the Poisson: Intuitive

Because we derived this by holding $np = \lambda$ constant:

- **Mean:** $E[X] = \lim_{n \to \infty}(np) = \lambda$
- **Variance:** $Var(X) = \lim_{n \to \infty} np(1 - p) = \lambda(1 - 0) = \lambda$

# Poisson Expectation: Direct Derivation

Let $X \sim \text{Poisson}(\lambda)$. By definition, the expected value is:

$$E[X] = \sum_{k=0}^{\infty} kP(X = k) = \sum_{k=0}^{\infty} k \frac{\lambda^k e^{-\lambda}}{k!}$$

The term for $k = 0$ is exactly zero, so we can start the sum at $k = 1$. We cancel $k$ with $k!$, leaving $(k-1)!$ in the denominator:

$$E[X] = \sum_{k=1}^{\infty} \frac{\lambda^k e^{-\lambda}}{(k-1)!}$$

We factor out $\lambda e^{-\lambda}$ to shift the index. Let $j = k - 1$:

$$E[X] = \lambda e^{-\lambda} \sum_{k=1}^{\infty} \frac{\lambda^{k-1}}{(k-1)!} = \lambda e^{-\lambda} \sum_{j=0}^{\infty} \frac{\lambda^j}{j!}$$

Recognizing the series expansion for $e^{\lambda} = \sum_{j=0}^{\infty} \frac{\lambda^j}{j!}$:

$$E[X] = \lambda e^{-\lambda}(e^{\lambda}) = \lambda \quad \blacksquare$$

# Poisson Variance: Second Factorial Moment

To find the variance, we first calculate the **second factorial moment**, $E[X(X-1)]$:

$$E[X(X-1)] = \sum_{k=0}^{\infty} k(k-1)\frac{\lambda^k e^{-\lambda}}{k!}$$

The terms for $k=0$ and $k=1$ are zero. Starting the sum at $k=2$, we cancel $k(k-1)$ with the $k!$ in the denominator:

$$E[X(X-1)] = \sum_{k=2}^{\infty} \frac{\lambda^k e^{-\lambda}}{(k-2)!}$$

Factor out $\lambda^2 e^{-\lambda}$ and let $j = k-2$:

$$E[X(X-1)] = \lambda^2 e^{-\lambda} \sum_{k=2}^{\infty} \frac{\lambda^{k-2}}{(k-2)!} = \lambda^2 e^{-\lambda} \sum_{j=0}^{\infty} \frac{\lambda^j}{j!}$$

Once again, the infinite sum resolves exactly to $e^{\lambda}$:

$$E[X(X-1)] = \lambda^2 e^{-\lambda}(e^{\lambda}) = \lambda^2$$

# Poisson Variance: Direct Derivation

We now have the two necessary components: $E[X(X-1)] = \lambda^2$ and $E[X] = \lambda$.

### 1. Find the Second Moment $E[X^2]$

Expand the factorial moment: $E[X(X-1)] = E[X^2] - E[X]$.

Rearranging gives us:

$$\begin{aligned} E[X^2] &= E[X(X-1)] + E[X] \\ &= \lambda^2 + \lambda \end{aligned}$$

### 2. Calculate Final Variance

Apply the standard variance formula $Var(X) = E[X^2] - (E[X])^2$:

$$\begin{aligned} Var(X) &= (\lambda^2 + \lambda) - (\lambda)^2 \\ &= \lambda^2 + \lambda - \lambda^2 \\ &= \lambda \quad \blacksquare \end{aligned}$$

# Waiting Time: The Geometric Distribution

We return to our sequence of independent Bernoulli trials (where $P(\text{Success}) = p$ and $P(\text{Failure}) = q = 1 - p$).
Instead of counting the *number* of successes in a fixed time, we now ask:
**How long must we wait until the very first success?**

## Defining the Random Variable

Let $X$ be the number of trials needed to get the first success.

- The only way for $X = k$ to occur is to have exactly $k - 1$ failures followed immediately by 1 success.
- Because the trials are independent, we multiply their probabilities.

## The Geometric PMF

We say $X \sim \text{Geometric}(p)$. Its Probability Mass Function is:

$$P(X = k) = q^{k-1}p, \quad \text{for } k = 1, 2, 3, \ldots$$

# Geometric Expectation:

By definition, $E[X] = \sum_{k=1}^{\infty} kP(X = k) = \sum_{k=1}^{\infty} kq^{k-1}p$.
We can pull the constant $p$ outside the sum:

$$E[X] = p \sum_{k=1}^{\infty} kq^{k-1}$$

To evaluate this infinite series, we recall the standard geometric series for $|q| < 1$:

$$\sum_{k=0}^{\infty} q^k = \frac{1}{1-q}$$

If we take the derivative of both sides with respect to $q$, the $k$ drops down:

$$\frac{d}{dq} \sum_{k=0}^{\infty} q^k = \sum_{k=1}^{\infty} kq^{k-1} = \frac{d}{dq}(1-q)^{-1} = \frac{1}{(1-q)^2}$$

Substituting this result back into our expectation formula:

$$E[X] = p\left[\frac{1}{(1-q)^2}\right] = p\left[\frac{1}{p^2}\right] = \frac{1}{p} \quad \blacksquare$$

# Geometric Variance and Intuition

To find the variance, we would use the second derivative of the geometric series to find the second factorial moment $E[X(X-1)]$.

### Moments Summary

For $X \sim \text{Geometric}(p)$:

- **Mean:** $E[X] = \frac{1}{p}$
- **Variance:** $Var(X) = \frac{1-p}{p^2}$

### Physical Intuition

The expected value $E[X] = 1/p$ aligns perfectly with human intuition:

- If you flip a fair coin ($p = 1/2$), you expect to wait $1/(1/2) = 2$ flips to see the first Heads.
- If you roll a fair die looking for a six ($p = 1/6$), you expect to wait $1/(1/6) = 6$ rolls to see the first 6.

# The Memoryless Property

A unique and defining characteristic of the Geometric distribution is that it is **memoryless**.

### Formal Definition

For any integers $s > 0$ and $t > 0$:

$$P(X > s + t \mid X > s) = P(X > t)$$

The probability of waiting $t$ additional trials for a success, given that we have already waited $s$ trials, is exactly the same as the probability of waiting $t$ trials from the very beginning.

The samples do not "remember" past failures. Suppose you are rolling a fair die waiting for a 6.

- You have already rolled 10 times and failed.
- The die is not "due" for a 6. The probability that it takes 3 *more* rolls is exactly the same as taking 3 rolls on a brand new die.

# Proof of the Memoryless Property

**Prerequisite:** Check that $P(X > k) = q^k$. (To fail to get a success in the first $k$ trials means getting $k$ failures in a row).

Using the definition of conditional probability:

$$P(X > s + t \mid X > s) = \frac{P(X > s + t \text{ and } X > s)}{P(X > s)}$$

Since the event $\{X > s + t\}$ is a subset of the event $\{X > s\}$, their intersection is simply $\{X > s + t\}$. We substitute our tail probabilities:

$$\begin{aligned} P(X > s + t \mid X > s) &= \frac{P(X > s + t)}{P(X > s)} \\ &= \frac{q^{s+t}}{q^s} \\ &= q^t \\ &= P(X > t) \quad \blacksquare \end{aligned}$$

**Conclusion:** The past history of failures algebraically cancels out.

# Exercise: Which Distribution?

Let's test our intuition. For each scenario, identify the most appropriate distribution model and its parameters.

### 1. The Single Seed

A plant seed has a 25% chance to sprout with green leaves. We plant exactly one seed. Let $X = 1$ if it is green, and $X = 0$ if it is not.

### 2. The Sales Calls

A worker makes 50 phone calls. Each call has an independent 5% chance of resulting in a sale. Let $X$ be the total number of sales made today.

### 3. The Random Inspection

A computer randomly picks one shipping box out of 12 to inspect. Every box is equally likely to be picked. Let $X$ be the chosen box number.

# Exercise: Which Distribution

Identify the most appropriate distribution model and its parameters.

### 4. Summer Storms

Historical weather data shows that a specific town gets hit by lightning an average of 14 times per summer. Let $X$ be the total number of lightning strikes this coming summer.

### 5. The Rare Blood Type

Only 2% of blood donors have a specific rare antibody. A doctor tests donated blood bags one by one. Let $X$ be the total number of bags tested until they find the **very first** match.

# Exercise

### 1. The Environmental Sensor

An oceanic sensor is designed to detect a specific microplastic particle. When a particle passes the sensor, there is a 10% chance the sensor successfully registers it.

- How many particles must pass by the sensor to ensure the probability of detecting **at least one** particle is strictly greater than 95%?

### 2. The Bakery Orders

A bakery receives an average of 2.5 large catering orders per week. Assume the weekly number of catering orders follows a Poisson distribution.

- **(a)** What is the probability the bakery receives at least 2 catering orders this week?
- **(b)** If we know the bakery received *at least one* order this week, what is the exact probability it received exactly 3 orders?

# Exercise

### 3. The Tabletop Game

In a role-playing game, a player rolls a fair 20-sided die. They need to roll a "Natural 20" (a probability of 1/20) to break a curse. They keep rolling until they succeed.

- What is the probability it takes **more than** 5 rolls to break the curse?

### 4. The Mystery Raffle

A charity raffle sells consecutively numbered tickets starting from Ticket #1 up to Ticket #$N$. The winning ticket is drawn with perfect fairness. If the **expected value** of the winning ticket number is 42:

- **(a)** How many total tickets ($N$) were sold?
- **(b)** What is the variance of the ticket numbers?

# From Discrete to Continuous

Until now, our random variables jumped between distinct, separate values (counts, dice rolls, iterations). Now, we enter the continuous realm, where a variable can take **any real number** within a range (like exact weight, continuous time, or distance).

## The Probability Density Function (PDF)

Instead of a Probability Mass Function (PMF), continuous variables are described by a PDF, denoted as $f(x)$.

- $f(x)$ is **not** a probability itself. It is a density.
- Probabilities are calculated by finding the **area under the curve** over a specific interval.

Because a continuous range has infinitely many points, the probability of hitting one exact, infinitely precise number is always zero:

$$P(X = c) = \int_c^c f(x)dx = 0$$

# The Continuous Uniform Distribution

We begin with the continuous equivalent of perfect fairness.

### The Definition

A random variable $X \sim \text{Uniform}(a, b)$ is equally likely to take any real value in the continuous interval $[a, b]$. Because the density is evenly spread out, the PDF is a perfectly flat, horizontal line.

### The PDF Formula

To ensure the total area under the curve equals exactly 1 (a rectangle of width $b - a$), the height must be $\frac{1}{b-a}$:

$$f(x) = \begin{cases} \frac{1}{b-a} & \text{for } a \leq x \leq b \\ 0 & \text{otherwise} \end{cases}$$

# Continuous Uniform: Expected Value

Let $X \sim \text{Uniform}(a, b)$ with PDF $f(x) = \frac{1}{b-a}$. We find the expected value via integration:

$$\begin{aligned} E[X] &= \int_a^b x f(x) dx \\ &= \int_a^b x \left( \frac{1}{b-a} \right) dx \\ &= \frac{1}{b-a} \left[ \frac{x^2}{2} \right]_a^b \\ &= \frac{1}{b-a} \left( \frac{b^2 - a^2}{2} \right) \end{aligned}$$

Factoring the difference of squares in the numerator, $b^2 - a^2 = (b-a)(b+a)$:

$$E[X] = \frac{1}{b-a} \frac{(b-a)(b+a)}{2} = \frac{a+b}{2} \quad \blacksquare$$

# Continuous Uniform: Variance

First, we calculate the second moment, $E[X^2]$:

$$E[X^2] = \int_a^b x^2 \left(\frac{1}{b-a}\right) dx = \frac{1}{b-a}\left[\frac{x^3}{3}\right]_a^b$$
$$= \frac{b^3 - a^3}{3(b-a)} = \frac{(b-a)(a^2+ab+b^2)}{3(b-a)} = \frac{a^2+ab+b^2}{3}$$

Now, apply the variance formula $Var(X) = E[X^2] - (E[X])^2$:

$$Var(X) = \frac{a^2+ab+b^2}{3} - \left(\frac{a+b}{2}\right)^2$$
$$= \frac{a^2+ab+b^2}{3} - \frac{a^2+2ab+b^2}{4}$$

Finding a common denominator of 12:

$$Var(X) = \frac{4a^2+4ab+4b^2-3a^2-6ab-3b^2}{12}$$
$$= \frac{a^2-2ab+b^2}{12} = \frac{(b-a)^2}{12} \quad \blacksquare$$

# Deriving the Exponential: The Setup

Assume we have a continuous random variable $X \geq 0$ that possesses the memoryless property:

$$P(X > s + t \mid X > s) = P(X > t) \quad \text{for all } s, t \geq 0$$

Using the definition of conditional probability:

$$\frac{P(X > s + t \text{ and } X > s)}{P(X > s)} = P(X > t)$$

Since $(X > s + t)$ is a strict subset of $(X > s)$, their intersection is just $(X > s + t)$:

$$\frac{P(X > s + t)}{P(X > s)} = P(X > t) \implies P(X > s + t) = P(X > s)P(X > t)$$

To make this easier to read, let us define the **tail probability** (or survival function) as $G(x) = P(X > x)$. We now have a strict functional equation:

$$G(s + t) = G(s)G(t)$$

# Deriving the Exponential: The Solution

We need to find a tail probability function $G(x)$ that satisfies our memoryless equation:

$$G(s+t) = G(s)G(t)$$

### Finding the Function

The standard algebraic function that converts addition in its input to multiplication in its output is the exponential function:

$$e^{\lambda(s+t)} = e^{\lambda s} \cdot e^{\lambda t}$$

Because $G(x) = P(X > x)$ represents a probability, it has strict boundaries:

- It must start at $G(0) = 1$.
- It must strictly decrease as the waiting time $x$ increases.

To make the function decay rather than grow infinitely, the exponent must be negative. Thus, for some rate parameter $\lambda > 0$:

$$G(x) = e^{-\lambda x}$$

# The Exponential Distribution

We just proved that the tail probability must be $P(X > x) = e^{-\lambda x}$.

## 1. Find the CDF

The Cumulative Distribution Function, $F(x) = P(X \leq x)$, is simply the complement of the tail probability:

$$F(x) = 1 - P(X > x) = 1 - e^{-\lambda x}$$

## 2. Find the PDF

By the Fundamental Theorem of Calculus, the Probability Density Function $f(x)$ is the derivative of the CDF:

$$\begin{aligned} f(x) &= \frac{d}{dx} F(x) \\ &= \frac{d}{dx} \left(1 - e^{-\lambda x}\right) \\ &= -e^{-\lambda x} \cdot (-\lambda) \\ &= \lambda e^{-\lambda x} \quad \blacksquare \end{aligned}$$

# Exponential Distribution: Mean Derivation

Let $X \sim \text{Exponential}(\lambda)$ with PDF $f(x) = \lambda e^{-\lambda x}$. By definition:

$$E[X] = \int_0^\infty x\lambda e^{-\lambda x} dx$$

We use Integration by Parts (IBP): $\int u\, dv = uv - \int v\, du$.

- Let $u = x \implies du = dx$
- Let $dv = \lambda e^{-\lambda x} dx \implies v = -e^{-\lambda x}$

$$\begin{aligned}
E[X] &= \left[-xe^{-\lambda x}\right]_0^\infty - \int_0^\infty (-e^{-\lambda x}) dx \\
&= (0 - 0) + \int_0^\infty e^{-\lambda x} dx \quad (\text{Note: } \lim_{x\to\infty} \frac{x}{e^{\lambda x}} = 0 \text{ by L'Hôpital's}) \\
&= \left[-\frac{1}{\lambda} e^{-\lambda x}\right]_0^\infty \\
&= 0 - \left(-\frac{1}{\lambda}\right) = \frac{1}{\lambda} \quad \blacksquare
\end{aligned}$$

# Exponential Distribution: Second Moment

To find the variance, we first need $E[X^2] = \int_0^\infty x^2 \lambda e^{-\lambda x} dx$. Using IBP again:

- Let $u = x^2 \implies du = 2x\,dx$
- Let $dv = \lambda e^{-\lambda x} dx \implies v = -e^{-\lambda x}$

$$\begin{aligned} E[X^2] &= \left[-x^2 e^{-\lambda x}\right]_0^\infty - \int_0^\infty (-e^{-\lambda x}) \cdot 2x\,dx \\ &= 0 + 2\int_0^\infty x e^{-\lambda x} dx \end{aligned}$$

Notice that the remaining integral is almost $E[X]$. We multiply and divide by $\lambda$:

$$\begin{aligned} E[X^2] &= \frac{2}{\lambda} \int_0^\infty x \lambda e^{-\lambda x} dx \\ &= \frac{2}{\lambda} \cdot E[X] = \frac{2}{\lambda} \cdot \frac{1}{\lambda} = \frac{2}{\lambda^2} \end{aligned}$$

# Exponential Distribution: Variance

Now we apply the variance formula:

$$Var(X) = E[X^2] - (E[X])^2$$

### The Final Calculation

Substituting our derived moments:

$$\begin{aligned} Var(X) &= \frac{2}{\lambda^2} - \left(\frac{1}{\lambda}\right)^2 \\ &= \frac{2}{\lambda^2} - \frac{1}{\lambda^2} \\ &= \frac{1}{\lambda^2} \quad \blacksquare \end{aligned}$$

### Summary for $X \sim \text{Exp}(\lambda)$

- **Mean:** $E[X] = 1/\lambda$
- **Standard Deviation:** $SD(X) = 1/\lambda$